

Hazard Analysis Flexin

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
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1 Introduction

This document aims to illustrate the hazards that may occur during the development and deployment of Flexin'. It defines hazards for this project, the purpose of analyzing them, and how the team will mitigate them to ensure safe use of Flexin'.

1.1 Flexin' Problem Statement

Personal finance and fitness follow the same pattern: the payoff for effort put in today shows up years later. Someone who invests \$100 a month from age 25, assuming a 7% annual return, contributes \$48,000 over 40 years and ends up with roughly \$262,000. Waiting until 35 to start the same plan leaves them with about \$122,000. The same logic applies to exercise. A single walk or workout changes very little, but the habit, kept up for years, is what separates a healthy 40-year-old from an unhealthy one. In both cases the early actions matter most, and in both cases they are the easiest to skip, because the reward is too far away to feel real.

The concepts behind this are not widely understood, and young people understand them the least. A 2012 survey of 6,911 Canadians asked three basic questions covering compound interest, inflation, and risk diversification [1]. Only 42% of respondents answered all three correctly. Among respondents under 35, that number fell to about 30%, the lowest of any age group, and 44% of them answered “don't know” to at least one question. Only about half of respondents under 35 knew that buying a single company's stock is usually riskier than buying a stock mutual fund. The same study found that respondents who answered all three questions correctly were 10 percentage points more likely to have retirement savings, which suggests that this gap in knowledge carries through to real financial outcomes.

Tools for practising each skill already exist, but separately. Stock market simulators let people trade with fake money, and fitness apps reward steps with points and badges. A market simulator only helps someone who is already motivated to learn about investing, which excludes most of the young adults the survey shows are furthest behind. Fitness rewards, meanwhile, rarely lead to anything beyond the app itself. No product currently connects the two so that a habit a user is already building (or wants to build) becomes their way into learning how money grows.

1.2 Hazard Analysis Introduction

A hazard as it pertains to Flexin' is a condition in the software that can create an unwanted effect of event on Flexin's users, or the general environment. In software, these hazards stem from poor design decisions or requirements. These hazards must be addressed prior to and during development in order to limit any unwanted impact on users.

As described above, Flexin' focuses on developing healthy habits for investing and fitness and teach users the effect of compounding. However, to accomplish this goal there are many hazards that may hurt users physically, psychologically, or financially. These hazards may come from poor software implementation, incompatible data, or faulty integration with outside sources. The hazard analysis document below aims to identify all hazards and how the team plans to address them.

2 Scope and Purpose of Hazard Analysis

[You should say what **loss** could be incurred because of the hazards. —SS]

3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

7 Roadmap

[Which safety requirements will be implemented as part of the capstone timeline? Which requirements will be implemented in the future? —SS]

8 Appendix — Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section. —SS]

8.1 AI Tools Used

[Give the name of the AI tool(s) used and the version. —SS]

8.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies. —SS]

[For each of the uses, identify which parts of the document were affected. —SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work —SS]

8.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

—SS]

9 Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

References

- [1] D. Boisclair, A. Lusardi, and P.-C. Michaud, "Financial literacy and retirement planning in Canada," *Journal of Pension Economics and Finance*, vol. 16, no. 3, pp. 277–296, 2017, doi: 10.1017/S1474747215000311.